Curriculum Vitae

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Sangwon Kim



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Research Interests

· Ultrafast tunneling in nanocontacts
· Strong field quantum optics
· Quasiclassical quantum theory
· Bohmian mechanics

Education

2019/02/25 - 2026/08/21 Ph.D. in Physics, Korea Advanced Institute of Science and Technology, Korea
· Advisor: Andrey S. Moskalenko
· Thesis: Quasiclassical theory of nonadiabatic tunneling in nanocontacts driven by classical and quantum light

2014/03/01 - 2019/02/24 B.S. in Physics, Korea Advanced Institute of Science and Technology, Korea

Professional Experience

2026/09/01 - 2027/08/31 Post-doctoral Researcher
TUQP, Korea Advanced Institute of Science and Technology, Korea

Awards/Appreciation

May 2024 Best Student Paper Award
ALTA 2024

Publication List

3. Tunneling driven by quantum light described via field Bohmian trajectories (Submitted)
Sangwon Kim, Seongjin Ahn, Denis V. Seletskiy, and Andrey S. Moskalenko
arXiv:quant-ph/2507.15972

2. Electronic Switching at the Atomic Spatio-Temporal Scale (Accepted)
Marwin Gedamke, Sebastian Grossenbach, Sven Väth, Matthias Falk, Stefan Eggert, **Sangwon Kim**, Andrey S. Moskalenko, Daniele Brida, Markus Ludwig, and Alfred Leitenstorfer
Nat. Nanotechnol.

1. Quasiclassical theory of non-adiabatic tunneling in nanocontacts induced by phase-controlled ultrashort light pulses
   Sangwon Kim, Tobias Schmude, Guido Burkard, and Andrey S. Moskalenko
   New. J. Phys. 23 083006 (2021). [arXiv:cond-mat.mes-hall/2011.15125]

Oral Presentations

2. QAMTS 2024 at Donostia-San Sebastián, Spain (June 16 - 21, 2024)
   Quasiclassical approach to dynamical aspects of tunneling in nanocontacts induced by ultrashort light pulses

1. OPC 2021 at Jeju, Korea (July 4 - 7, 2021)
   Quasiclassical theory of non-adiabatic tunneling in nanocontacts induced by ultrashort light pulses

Poster Presentation

7. UP 2026
   Sapporo, Japan (August 2 - 7, 2026)

6. ALTA 2026
   Jeju, Korea (May 6 - 9, 2026)

5. ALTA 2024
   Jeju, Korea (May 1 - 4, 2024)

4. GRC Ultrafast Phenomena in Cooperative Systems
   Lucca (Barga), Italy (February 4 - 9, 2024)

3. ALTA 2023
   Jeju, Korea (May 1 - 4, 2003)

2. ICAMD 2021
   Jeju, Korea (December 6 - 10, 2021)

1. GRS/GRC Ultrafast Phenomena in Cooperative Systems
   Lucca (Barga), Italy (February 1 - 7, 2020)

Teachings (Lecturing Experience)

1. Substitute Lecturer, General Physics II, KAIST, Fall 2026
   · Delivered four full lecture sessions on behalf of the course instructor.

Teachings (Teaching Assistantships)

14. TA, Quantum Optics, KAIST, Spring 2026

13. Vice Head TA, General Physics I, KAIST, Fall 2025

12. TA, General Physics II, KAIST, Spring 2025

11. TA, General Physics II, KAIST, Fall 2024

10. TA, Graduate Quantum Mechanics II, KAIST, Fall 2023

9. TA, Graduate Classical Mechanics, KAIST, Spring 2023

8. TA, Physics Experiment I, KAIST, Fall 2022

7. TA, Quantum Optics, KAIST, Spring 2022

6. TA, Nonlinear Optics, KAIST, Fall 2021

5. TA, Graduate Classical Mechanics, KAIST, Spring 2021
4. TA, Classical Mechanics II, KAIST, Fall 2020
3. TA, Classical Mechanics I, KAIST, Spring 2020
2. TA, Experiment Oriented General Physics II, KAIST, Fall 2019
1. TA, General Physics II, KAIST, Spring 2019

Other Activities

2. Visited femtoQ, Prof. Denis V. Seletskiy Group, Polytechnique Montréal
   Montréal, Canada (December 5 - December 20, 2025).
1. Visited Prof. Guido Burkard Group and Prof. Alfred Leitenstorfer Group, Universität Konstanz
   Konstanz, Germany(January 7 - February 17, 2020).

Skills

1. Professional working proficiency in English.
2. Proficient in Mathematica, Maple, and Python/Jupyter.
3. Experienced with Linux Bash and C.
4. Completed all coursework required for a Bachelor's degree in Electrical Engineering, although the degree was not formally awarded.
5. Fully audited (participated all course activies including homeworks, quizzes and exams) the following courses:
   · Modern Algebra I (MAS311, Department of Mathematical Sciences, KAIST)
   · Introduction to Differential Geometry (MAS321, Department of Mathematical Sciences, KAIST)
   · Complex Function Theory (MAS541, Department of Mathematical Sciences, KAIST)